Department	**Department of CSE [IoT & Cybersecurity Including Blockchain Technology]**		
Subject Name	**Foundation of Blockchain Technology**	**Subject Code**	**IS-501**

Type of Question- 3 Marks

Q.No	**Question**	**Blooms Level**	**CO**
1	Differentiate between public and private blockchain.	L2	1
2	What is cryptographic hash function? Write the properties of hash function.	L2	3
3	Explain the application of SHA 256 algorithm in digital signing verification in detail	L3	3
4	What is the main purpose of the nonce in PoW?	L2	1
5	A block contains 4 transactions: T1, T2, T3, T4. The Merkle root is stored in the header. a. If T3 changes, explain what happens to the Merkle root and the block hash. b. Why does this mechanism ensure efficient verification and tamper-proofing?	L3	5
6	A blockchain network has validators holding a total of 900 ETH stake. At one checkpoint, 580 ETH worth of votes are recorded, and at the next checkpoint, 620 ETH worth of votes are recorded. The network finalizes a checkpoint only if a sufficient fraction of the total stake has supported the link between consecutive checkpoints. Determine whether the second checkpoint will be finalized, and if not, how much additional stake is required. Explain the concept involved in brief.	L3	1
7	A blockchain network requires miners to solve a hash puzzle with a difficulty target of 0000FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF. Miner A finds a valid nonce after 45 seconds, Miner B finds a valid nonce after 60 seconds, and Miner C finds a valid nonce after 30 seconds. The network accepts the first valid solution it receives to add a new block. 1. Which miner's block will be added to the blockchain? 2. If the network wants blocks to be mined roughly every 1 minute, should it increase or decrease the difficulty based on the observed times?	L3	1
8	Find any five valid points on the elliptic curve: $y^2 = x^3 + 2x + 3 \ (mod\ 17)$ Show the necessary calculations	L3	3
9	Differentiate between cold and hot wallets	L2	4
10	What do you understand by Turing completeness of smart contract?	L2	4
11	What do you mean by re-entrancy attack? Explain with the help of example.	L2	4

12	What are Solidity data types? Explain any three commonly used data types.	L2	4
13	Write a simple Solidity function to check whether a number is even or odd.	L3	4
14	Define a smart contract and explain its major characteristics.	L2	4
15	Differentiate between traditional contracts and smart contracts	L2	4
16	Given a smart contract that transfers funds before updating the user balance, identify the possible vulnerability and explain it.	L3	4
17	Explain the concepts of front running attacks in blockchain.	L2	4
18	What is a 51% attack? Explain how it can facilitate double spending.	L3	5
19	Define Zero-Knowledge Proof (ZKP) and explain its basic concept.	L1	2
20	Define mining difficulty in blockchain and explain its importance.	L2	5
21	A network consists of 10 nodes, out of which 7 are honest and 3 are Byzantine. Determine whether it satisfies the classical BFT requirement. Justify your answer.	L3	2
22	Explain the significance of the condition $n \geq 3f + 1$ in classical Byzantine Fault Tolerant consensus protocols.	L2	2
23	Differentiate between Bitcoin and Ethereum.	L4	2
24	Explain the concept of Gas and Gas Limit in Ethereum.	L2	4
25	Illustrate how Ethereum executes a smart contract when a user submits a transaction.	L2	4
26	What is consensus in Bitcoin? Explain its importance.	L2	5
27	Define Altcoin and explain its significance in the cryptocurrency ecosystem.	L2	5
28	Define anonymity in blockchain technology.	L1	5
29	Define a fork in blockchain technology and explain why forks occur.	L2	5
30	Give suitable examples of Hard Forks in blockchain technology.	L2	5

Type of Question- 5 Marks

Q.No	Question	Blooms Level	CO
1	Examine the working principal of Smart Contracts and discuss the applications of Smart Contracts and write advantage and disadvantages of Smart Contracts.	L4	4
2	Discuss Block Chain Technology with help of architecture and explain major characteristics of Block chain Technology.	L2	1
3	With a neat diagram explain the block header design.	L2	1
4	What are Merkle trees? How important are Merkle trees in the Block chain?	L2	5
5	Discuss in detail about the Cryptographic Hash Function and Properties of Hash Function.	L2	3
6	In SHA-256, after completing 64 rounds for a message block, each working variable is added to its corresponding current hash-state word modulo 2^{32}. Consider the following values after the 64 rounds.	L3	3

	Hash-state word / **Current hash state** / **Working variable** H_0 4,294,967,290 15 H_1 100 250 H_2 4,294,967,280 30 H_3 500 200 H_4 1,000 2,000 H_5 4,294,967,295 1 H_6 750 250 H_7 2,000 3,000 Calculate all eight updated hash-state words using modular addition. Then determine the total digest length in bits and the number of hexadecimal characters in the final SHA-256 digest. ($2^{32} = 4{,}294{,}967{,}296$)		
7	Given the elliptic curve $y^2 = x^3 + 2x + 2$ $(mod\ 17)$, calculate the coordinates of P+Q, where P=(5,1) and Q= (6,3).	L3	3
8	Explain the complete lifecycle of a smart contract, from development to deployment and execution.	L2	4
9	Explain re-entrancy, integer overflow and denial-of-service attacks with suitable examples.	L4	4
10	Analyze the differences between Ethereum and Bitcoin in terms of architecture, consensus and applications.	L4	4
11	Evaluate the transition of Ethereum from Proof of Work to Proof of Stake in terms of energy efficiency, security and scalability.	L5	1
12	A patient's medical records are stored in a centralized hospital database. Analyze the limitations of this approach and explain how blockchain can overcome them.	L5	5
13	A user loses access to a Bitcoin wallet but has securely stored the recovery seed phrase. Analyze how the wallet can be recovered and identify the associated security risks.	L4	5
14	A university wants to maintain student records (mark sheets, certificates) in a tamper proof manner. Design a small block chain based solution. Explain what information will go inside a block header and how consensus will ensure fairness.	L6	1
15	Analyze the role of the previous block hash, Merkle root, timestamp, nonce, and difficulty target in a block header.	L4	1
16	Illustrate how miners compete to generate a valid block using the Proof of Work mechanism.	L3	1
17	Illustrate the working of a hybrid consensus model that combines Proof of Work and Proof of Stake.	L3	1
18	A miner performs 10^6hash attempts per second. Calculate the number of attempts performed in 10 minutes.	L3	1
19	Evaluate the relationship between mining difficulty, computational power, block generation time, and network security.	L4	1

20	Construct a Merkle Tree for four transactions T_1, T_2 , T_3 and T_4showing the intermediate hashes and the final Merkle Root	L3	5
21	Illustrate the execution flow of a smart contract when a transaction invokes one of its functions.	L2	4
22	Explain the process of generating and verifying a digital signature using public-key cryptography.	L3	3
23	Analyze the role of cryptographic hashing in block linking and transaction integrity.	L4	3
24	Differentiate between Proof of Work and Proof of Stake in terms of block validation, resource requirements, and security.	L4	1
25	Alice and Bob use ECC for secure key exchange. Given G=(5,1) , Alice's private key a=5, and b=7 , calculate both public keys and verify that both parties obtained the same shared secret . use $x^3 + 2x + 2 \bmod(17)$	L3	3
26	Analyze the difference between public, private, internal and external visibility specifiers with suitable examples.	L2	4
27	Design a simple smart contract for an automated voting system using Solidity.	L6	4
28	Explain the concept of distributed ledger technology (DLT). Differentiate between traditional databases and distributed ledger technology with suitable examples.	L4	1
29	Explain symmetric and asymmetric key cryptography. Differentiate between them based on key management, security and applications in blockchain.	L4	3
30	Describe the different types of blockchain networks: Public, Private, Consortium and Hybrid Blockchain. Compare their features, advantages and limitations.	L4	1
31	Explain the concept of decentralization in blockchain. How does decentralization eliminate the need for a trusted central authority?	L2	1
32	What is a fork in blockchain? Explain the differences between soft fork and hard fork with suitable examples.	L4	5
33	Explain the role of nodes in a blockchain network. Discuss the functions of full nodes, light nodes and mining nodes.	L2	1
34	What is a digital certificate? Explain the role of Public Key Infrastructure (PKI) and Certificate Authorities in establishing digital trust.	L2	3
35	Explain the concept of a 51% attack in blockchain. Discuss how such an attack can affect network operations and suggest suitable preventive measures.	L5	5
36	What is a replay attack in blockchain? Explain how it occurs and discuss the techniques used to prevent replay attacks.	L4	4
37	Explain the working of a Bitcoin transaction with a suitable example. Discuss the role of inputs, outputs, transaction fees and change addresses.	L3	5
38	What is the Unspent Transaction Output (UTXO) model? Explain its working with a suitable numerical example.	L3	5
39	Explain the concept of Bitcoin mining rewards and transaction fees. How does Bitcoin control the supply of new bitcoins?	L4	5
40	What is a cryptocurrency wallet? Explain the different types of wallets,	L3	5

	including hot wallets, cold wallets, custodial wallets and non-custodial wallets.		

41	Consider the following Solidity smart contract:
	// SPDX-License-Identifier: MIT
	pragma solidity ^0.8.21;
	
	contract Counter {
	uint public count = 5;
	
	function increment() public {
	count = count + 2;
	}
	
	function decrement() public {
	count = count - 1;
	}
	
	function reset() public {
	count = 0;
	}
	}
	
	Answer the following:
	
	a) What is the initial value of `count`?
	
	b) Determine the value of count after executing the following sequence:
	increment();
	increment();
	decrement();
	increment();
	c) What will be the output of calling count() after these operations?
	d) Explain the purpose of the public keyword in the state variable declaration.
		L3	4

42	Study the following Solidity program:
	// SPDX-License-Identifier: MIT
	pragma solidity ^0.8.21;
	
	contract StudentMarks {
	function calculateGrade(uint marks)
	public pure returns (string memory)
	{
	if (marks >= 80) {
	return "A";
	}
		L3	4

	else if (marks >= 60) {
	return "B";
	}
	else if (marks >= 40) {
	return "C";
	}
	else {
	return "Fail";
	}
	}
	
	function totalMarks() public pure returns (uint) {
	uint total = 0;
	
	for (uint i = 1; i <= 5; i++) {
	total = total + (i * 10);
	}
	
	return total;
	}
	}
	
	Answer the following:
	
	a) What will be the output of `calculateGrade(75)`?
	
	b) What will be the output of `calculateGrade(35)`?
	
	c) Calculate the output of `totalMarks()`.
	
	d) Explain how the `for` loop works in this program.		

43	Consider the following Solidity smart contract and answer the questions given below:
	// SPDX-License-Identifier: MIT
	pragma solidity 0.8.34;
	
	contract CompareStrings {
	
	string public String = "Hello World";
	
	function setString(string memory _String) public {
	String = _String;
	}
	
	function compareTwoStrings(string memory _myString)
		L4	4

	public view returns(bool) {
	
	return String == _myString;
	}
	}
	
	Answer the following questions:
	
	a) What is the initial value of the state variable `String` after deploying the contract?
	
	b) What will be the output of `compareTwoStrings("Hello World")` immediately after deployment?
	
	c) What will happen when the following function calls are executed in sequence?
	
	setString("Blockchain Technology");
	compareTwoStrings("Hello World");
	
	d) What will be the output of the following function call?
	compareTwoStrings("Blockchain Technology");
	
	e) Identify the error in the string comparison statement and write the correct Solidity statement to compare two strings.		

44	What is a Decentralized Application (DApp)? Explain its architecture and differentiate between DApps and traditional web applications.	L4	4
45	Explain the concept of blockchain interoperability. Discuss the importance of cross-chain communication and the challenges involved in connecting different blockchain networks.	L2	1
46	Explain the structure of a Solidity program. Discuss the role of state variables, functions, constructors, events and modifiers with suitable examples.	L2	4
47	What is gas in Ethereum? Explain gas limit, gas price and transaction fees with a suitable numerical example.	L3	4
48	Discuss the role of blockchain in digital identity management. Explain how blockchain can support secure identity verification and give suitable real-world applications.	L4	5
49	Explain the applications of blockchain in supply chain management. How can blockchain improve product traceability, transparency and accountability?	L4	5

50	Study the following Solidity program:
	// SPDX-License-Identifier: MIT
	pragma solidity ^0.8.21;
	
	contract StudentRecord {
		L3	4

	mapping(uint => uint) public marks;
	
	constructor() {
	marks[101] = 75;
	marks[102] = 85;
	}
	
	function updateMarks(uint id, uint newMarks)
	public
	{
	marks[id] = newMarks;
	}
	
	function addBonus(uint id, uint bonus)
	public
	{
	marks[id] = marks[id] + bonus;
	}
	}
	
	Answer the following:
	
	a) What values are assigned to student IDs 101 and 102 when the contract is deployed?
	
	b) Determine the marks of student 101 after executing:
	
	updateMarks(101, 80);
	addBonus(101, 5);
	
	c) What will be the output of `marks(102)`?
	
	d) What will happen if `marks(103)` is called without previously assigning any value?
	
	e) Explain the role of the constructor and mapping in this smart contract.		

Type of Question- Read and Observer --15 Marks

Q.No	Question	Blooms Level	CO
1	Pharmaceutical Supply Chain Management A pharmaceutical company wants to track medicines from manufacturing to delivery. Currently, manufacturers, distributors and retailers maintain separate databases. This	L5	5

	creates difficulties in tracking medicine movement, identifying counterfeit products and verifying product authenticity. (a) Identify the limitations of the existing centralized record-management system. (b) Design a blockchain-based solution for tracking medicines from manufacturing to final delivery. Explain how blocks, transactions and participating nodes will work in this system. (c) Analyze how blockchain improves transparency, traceability and accountability in pharmaceutical supply chains. Also mention its limitations.		
2	A financial organization stores customer information and transaction records on a centralized server. Due to a cyberattack, the server becomes unavailable, and the organization faces difficulties in recovering accurate transaction records. The management decides to explore blockchain technology. (a) Explain the limitations of centralized database architecture in terms of security, availability and data integrity. (b) Explain how distributed ledger technology and decentralization can address these limitations. Support your answer with a suitable architecture diagram. (c) Evaluate the advantages and disadvantages of implementing blockchain in the organization's record-management system.	L4	1
3	University Certificate Verification System A university wants to develop a blockchain-based system to maintain student certificates and academic records. Employers should be able to verify certificates without repeatedly contacting the university. The system must ensure that certificates cannot be secretly modified. (a) Explain how cryptographic hash functions can be used to generate unique digital fingerprints of student certificates. (b) Design a blockchain-based certificate verification system. Explain the role of transactions, block headers, previous block hashes and Merkle roots. (c) Explain the complete certificate verification process and analyze how the system detects unauthorized modifications.	L4	3
4	Smart Contract-Based Financial Transactions An organization develops a Solidity smart contract to transfer funds automatically when predefined conditions are satisfied. During execution, a programming error causes incorrect fund transfers. The organization wants to improve the security and reliability of its smart contracts. (a) Explain the complete lifecycle of a smart contract, from development and compilation to deployment and execution. (b) Discuss any three common smart contract vulnerabilities with suitable examples and their preventive measures. (c) Explain the importance of testing, debugging and security auditing before deploying a smart contract on Ethereum.	L5	4
5	Ethereum-Based Online Payment System An online business accepts payments through Ethereum. Customers initiate transactions to purchase	L4	4

	products, but some transactions remain pending due to insufficient gas fees. The company also wants to understand the security and transaction-processing mechanism of Ethereum. (a) Explain the role of gas, gas limit and gas price in Ethereum transactions. Illustrate with a suitable numerical example. (b) Explain how an Ethereum transaction is processed, validated and recorded on the blockchain. (c) Compare Ethereum and Bitcoin in terms of architecture, consensus mechanism, transaction processing and applications.		
6	Case Study: Interoperability Between Blockchain Networks Two financial organizations maintain separate blockchain networks. One organization uses a permissioned blockchain, while the other uses a public blockchain. They want to exchange transaction information securely without maintaining a common centralized database. (a) Explain blockchain interoperability and why it is required for communication between different blockchain networks. (b) Discuss the working of cross-chain communication and explain the role of bridges or interoperability protocols. (c) Analyze the security challenges associated with blockchain interoperability and suggest suitable measures to overcome them.	L4	1